

whose output is appropriate for grid display is applied, or when importing certain types of annotation data. For an instantaneous display of y-axis values of a 3D plot, we can invoke a slice layer. As Sonic Visualiser website puts it, "a slice is to a colour 3D plot as a spectrum is to a spectrogram."

Plugins to generate new layers and create effects

Third-party plugins can be used with Sonic Visualiser to enhance understanding of the input signal. The general format for audio-analysis and feature-extraction plugins is Vamp, which takes in audio input and generates output compatible with display of Sonic Visualiser layers. We could also have plugins in Linux Audio Developer's Simple Plugin API (LADSPA) or Disposable Soft Synth Interface (DSSI) formats to create effects like compression or to generate data using controlling parameters like sine-wave tone or white and pink noise.

Other simple, yet indispensable features

A measure tool helps us select the exact range of the signal we want to work with, or maybe observe repeatedly to detect something amiss. A friendly and appealing user interface lets us customise everything from default frequency to background colour.

Sonic Visualiser is structured around panes and layers. Panes are horizontally-scrollable areas that can be stacked one above the other, and a pane can have any number of layers in it. Easy export and import of audio files using export layer and handy annotation add to the glitz.

A mini iPod for signal processing

Processing a signal requires revisiting segments repeatedly, making changes to portions and studying results closely. Sonic Visualiser comes to our aid by offering a hassle-free

playback-and-pause platform. Much like a media player, we can use keyboard shortcuts to control the signal being viewed. It is also easy to align different waveforms along the same time scale and instant. This comes in very handy while comparing signals to find differences between similar-looking ones or when viewing multiple files concurrently. We can even time-stretch playback, slowing it down to as little as ten per cent of the original speed, while retaining a synchronised display.

Version 2.5 released in October 2015 works on Pentium4 and AMD64 processors. Developed at Centre for Digital Music, Queen Mary, University of London, and partially funded by many organisations, this month's DVD gives you the package for easy download. Let us begin our experiments with audio signals. **EFY**

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